

Designing the Return

HMI Design for Level 3 Autonomous-to-Manual Handover Transitions

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Abstract

Level 3 autonomous vehicles permit drivers to fully disengage from the driving task and engage in non-driving related activities. When the vehicle reaches the boundary of its operational design domain, it issues a Takeover Request (TOR), requiring the driver to resume full manual control within four to ten seconds. Current HMI design treats this as an alarm event: a visual alert, an acoustic signal, and in some systems a haptic cue, all firing simultaneously at the moment of handover. The research literature consistently shows this approach is insufficient. Drivers engaged in screen-based secondary tasks are cognitively and physically displaced from the driving state, and a sudden alarm does not rebuild the situation awareness required for safe takeover. This paper argues that the handover is not an alarm problem but a *transition experience design problem*, and that the real design opportunity lies in the 30 to 60 seconds preceding the TOR rather than in the alert moment itself. Drawing on the human factors, automotive HCI, and cognitive psychology literature, as well as a review of production HMI systems, this paper presents a framework for a screen-based interaction system that progressively reintroduces driving context, rebuilds situation awareness, and guides the driver toward readiness before the alert fires. The paper identifies three persistent gaps in current HMI research and practice, and proposes a five-state interface architecture as a foundation for future design and evaluation.

Introduction

Level 3 (L3) conditionally automated driving systems represent a fundamental shift in the human-machine relationship within the vehicle. Unlike Level 2, which requires continuous driver supervision, L3 systems legally permit the driver to disengage from the dynamic driving task entirely [29]. The driver may look away from the road, engage in non-driving related tasks (NDRTs), and devote their attention to activities previously impossible behind the wheel.

Mercedes-Benz DRIVE PILOT, the first L3 system to receive legal certification in the

United States, formalises this by unlocking entertainment on the central display when autonomous mode is active: drivers may play games, watch videos, or browse the internet while the vehicle manages navigation [20]. The system is marketed explicitly on giving drivers back “the most valuable asset: time” [18].

When the system’s operational capability reaches its limit, the driver must return from this mentally absent state to full situational awareness and manual control within four to ten seconds [10]. The mechanism for this transition in current production systems is a simultaneous alarm: a visual alert, an acoustic warning, and in some implementations a haptic signal [20, 18]. The driver is expected to shift from one cognitive state to another in response to a sudden shock.

A consistent finding across a decade of takeover research is that drivers engaged in NDRTs show significantly degraded takeover quality compared to drivers in an attentive monitoring state [37, 23, 17]. Response times increase, steering quality degrades, and the probability of unsafe maneuvers rises [38, 33]. The degradation scales with the duration of autonomous mode and the cognitive depth of the secondary task [12, 25].

Despite this evidence, design effort directed at the handover experience remains focused almost entirely on the alert itself: its modality, timing, and urgency calibration [34, 24, 26]. The period preceding the alert, what this paper terms the *pre-handover window*, is largely unaddressed in both research and production HMI.

The central design question this paper addresses is: *How do you design the in-car screen experience for the 60 seconds before and 10 seconds during a Level 3 handover, so that the driver arrives at the takeover moment prepared rather than startled?*

The paper proceeds as follows. Section 2 situates the design gap within the context of current production HMI. Section 3 reviews the literature on driver behavior during autonomous mode. Section 4 examines alert design research, with particular attention to pre-alert findings. Section 5 addresses the taxonomy of handover triggers and their implications for interface timing. Section 6 covers relevant emerging sensing technologies. Section 7 addresses ethical and regulatory dimensions of the design space. Section 8 synthesizes the literature into three persistent gaps and proposes a five-state interface architecture. Section 9 concludes with directions for future empirical work.

Background: Production HMI and the Pre-Handover Gap

The SAE taxonomy defines Level 3 as the point at which the vehicle accepts full operational responsibility within its certified operational design domain (ODD) and the driver is legally permitted to disengage entirely [29]. This has a direct consequence for HMI: an L3 interface must bring a driver back from a disengaged state reliably and safely, every

time, with no guarantee of what state the driver will be in when the request arrives.

Current Production HMI

DRIVE PILOT’s handover HMI consists of a visual alert on the central display and instrument cluster, an acoustic warning, and haptic feedback through the steering wheel. If the driver does not respond, the system executes a minimal-risk maneuver [18]. There is no designed interface experience between the entertainment unlock state and the TOR alert. The driver transitions from watching a film to receiving a full alarm with no intermediate step.

BMW’s Level 3 system for the 7 Series operated similarly, using audible and visual alerts with gradual braking as a fallback. BMW announced a retreat from Level 3 for its Neue Klasse lineup in February 2026 [6], citing sensor cost and the limited operational envelope of current approvals. This is a relevant market signal, though it does not change the design problem for the systems already deployed and certified.

Table 1 summarizes handover HMI across the main production systems.

Table 1. Handover HMI characteristics across production systems.

System	Level	NDRT screen mode	Pre-handover UX	Handover alert
Mercedes DRIVE PILOT	L3	Video, games, internet unlocked on central display	None designed before TOR	Visual + acoustic + wheel haptic
BMW Personal Pilot	L2++ (2026)	Ambient information only	Gradual braking fallback	Audible + visual
Tesla Full Self-Driving	L2	No; driver must supervise at all times	Not applicable	Torque check + visual prompt
Waymo (robotaxi)	L4	Full entertainment, no driver seat	Not applicable	None required

The pre-handover window is unaddressed across all current L3 production systems.

The pattern is consistent. The period from approximately 60 seconds before the TOR to the moment it fires is entirely undesigned in all current L3 implementations. This is not a hardware limitation; the screens, sensors, and processing infrastructure are present. It

is a design gap.

Driver Behavior During Autonomous Mode

Non-Driving Related Task Engagement

NDRT engagement increases significantly with the level of driving automation [36]. In L3 contexts, participants engaged with secondary tasks more frequently and at greater depth than in manual or L2 conditions [27, 35]. The most frequently observed tasks are screen-based: watching video, browsing the internet, typing [3]. All of them direct gaze entirely away from the road environment.

Van der Heiden et al. quantified the scale of this visual disengagement: in autonomous mode with a secondary screen task, drivers looked at the road only 6% of the time during the pre-alert period, compared to 71% in single-task conditions [33]. That 6% figure is the baseline state from which a handover interface must recover the driver.

Posture and Physical Displacement

Beyond gaze, NDRT engagement produces significant physical changes that directly affect takeover capability. Zhao et al. examined 54 drivers across 21 non-driving postures and found systematic patterns during autonomous mode: drivers recline, turn toward secondary screens, rest their heads on their hands, and remove their feet from the pedal area [38]. Foot position had the largest effect on braking reaction time; resting the chin on the hand had the largest effect on steering reaction time. A complementary study in Accident Analysis and Prevention confirmed that non-driving postures significantly impaired takeover performance across both reaction time and control quality measures [30].

The implication for interface design is concrete. A handover sequence cannot simply redirect the driver's gaze toward the road. It must also initiate physical reorientation toward the driving position, and the earlier that process begins, the better.

The Out-of-the-Loop Effect

Research published in Heliyon (2024) confirmed that the duration of autonomous mode negatively affects takeover performance: the longer the session, the more degraded situation awareness becomes, and the harder the cognitive re-entry into the driving task [12]. This out-of-the-loop effect is well-documented across multiple studies [22, 31]. A driver in autonomous mode for three minutes has a meaningfully different recovery requirement than one who has been in it for thirty minutes, and an effective pre-handover system should scale the transition window accordingly rather than treating all sessions as equivalent.

Task Switching and Cognitive Residue

Strayer et al. demonstrated that distraction effects from secondary cognitive tasks can persist for up to 27 seconds after the task has ended, decaying exponentially [32]. Van der Heiden et al. used this finding to justify a 20-second pre-alert window: 20 seconds is the minimum interval needed to begin recovering from secondary task distraction and reorienting to the driving environment [33].

Research on task-switching in L3 contexts found that drivers engaged in multiple NDRTs, or switching frequently between tasks, showed significantly worse takeover performance than those in a single sustained task [25]. The interface must not simply interrupt the secondary task but must actively manage the cognitive transition out of it. The distinction matters: stopping is not the same as recovering.

Visual Assistance and Situation Awareness Restoration

A 2024 study in Human Factors examined the impact of visual assistance on situation awareness during L3 takeover events [14]. Drivers provided with progressive visual guidance showing road context before the TOR fired showed meaningfully improved situation awareness at takeover, with effects on both reaction quality and downstream driving performance. This is among the most directly relevant findings for the design approach proposed in this paper, providing empirical support for screen-based pre-handover information delivery as a viable intervention.

Alert Design: From Alarms to Pre-Alerts

Multimodal Alert Effectiveness

Unimodal visual alerts are ineffective when drivers are engaged in secondary tasks [26, 34]. A driver watching video content on the central display will not reliably notice a visual alert on the same screen. Multimodal alerts combining visual, auditory, and haptic channels yield the shortest reaction times and the highest takeover quality [28]. Politis et al. found performance was worst with visual-only alerts, intermediate with visual-auditory combinations, and best with all three modalities together [26]. These findings argue for retaining audio and haptic channels in the final alert regardless of what precedes it in the transition sequence.

Pre-Alerts: The Case for Early Warning

The most directly relevant empirical evidence for the pre-handover approach comes from van der Heiden et al. [33]. Their simulator study with 24 participants tested auditory pre-alerts issued 20 seconds before the final handover warning. Two formats were compared: a repeated burst (three bursts of three beeps at 20, 10, and 1 seconds before the warning)

and an increasing pulse (continuous beeps with decreasing inter-stimulus interval from 1000ms to 50ms over 20 seconds).

The results were substantial. In dual-task conditions, pre-alerts caused drivers to look at the road 49–54% of the time during the pre-alert phase, compared to 23% without a pre-alert. Drivers disengaged from secondary tasks significantly earlier; the increasing pulse produced the fastest disengagement, approximately twice as fast as in the no pre-alert condition. Post-handover reaction times were faster, and speed reduction was earlier. Heart rate data added a physiological dimension: heart rate was significantly higher in the no pre-alert condition, suggesting that a sudden alarm under dual-task conditions produces a measurably elevated stress state at precisely the moment when the driver most needs to be calm.

The optimal pre-alert window is not simply “as early as possible.” A pre-alert issued too far in advance loses urgency and may be ignored; one issued too late provides insufficient preparation time [33]. The 20-second window in that study was calibrated to Strayer et al.’s cognitive residue finding. Notably, the increasing pulse format outperformed the repeated burst on most objective metrics but was rated as more annoying by roughly half the participants, who preferred the burst because it gave them a sense of finishing the secondary task before attending to the alert. This tension between effectiveness and subjective acceptability is a core design challenge for any pre-handover system.

Van der Heiden et al. explicitly called for future work on dynamic pre-alert timing that adapts to distraction level, task complexity, and trigger type [33]. That call remains largely unanswered in the literature.

Encoding Information in the Pre-Alert

Van der Heiden et al. also raise the possibility of encoding richer information in the pre-alert itself: not just “handover is coming” but why it is coming and what the driver should do [33]. Their study used only audio beeps, and the communicative potential of a visual pre-alert sequence on the central display was not explored. That unexplored space is the design territory this paper addresses.

Handover Triggers and Their UX Implications

Understanding what technically causes a TOR is necessary for designing a pre-handover interface. Different trigger types provide different amounts of preparation time and require different interface responses, so the architecture must account for the full range.

Taxonomy of Trigger Types

TORs can be classified along a predictability axis with three broad categories.

Planned and predictable triggers (30–60 seconds warning typically available): These include approach to the boundary of a certified highway segment as defined by the vehicle’s HD map and GPS geofence, anticipated speed threshold changes, and scheduled transitions at tolls or tunnels. Because the vehicle’s route planning system knows these boundaries in advance, a full pre-handover sequence can in principle begin well before the TOR fires.

Semi-predictable triggers (10–30 seconds): These include degradation of road markings, deteriorating weather or visibility conditions, construction zone detection, and emergency vehicle identification via acoustic or visual sensors. DRIVE PILOT uses dedicated microphones to detect emergency vehicle signals and a wheel well wetness sensor for precipitation [18]. These conditions develop progressively, providing a partial warning window.

Abrupt and unplanned triggers (less than 10 seconds): These include sudden sensor failure or occlusion, unexpected road geometry, or a confidence score falling below the safety threshold. DRIVE PILOT’s redundant sensor architecture is designed to convert single-sensor failures into semi-predictable handovers rather than abrupt losses of control [18], but abrupt triggers remain possible in real-world deployment.

Implications for Interface Architecture

An interface that assumes a fixed preparation window will be miscalibrated for most real-world handovers. The architecture must define distinct behavioral modes for each trigger category: a full pre-handover sequence for planned triggers, a compressed sequence for semi-predictable triggers, and a direct alert for abrupt ones, while still ensuring that the abrupt case does not feel more shocking than the constraints require. This adaptive timing requirement aligns with the call from van der Heiden et al. for dynamic pre-alert design [33] and with the broader literature on context-sensitive interruption management [2].

Emerging Technologies in the Design Space

Cognitive State Sensing

A significant limitation of current pre-handover design is that the vehicle has no direct read on the driver’s cognitive state. It knows the driver is watching a video; it does not know whether they are alert, fatigued, or cognitively overloaded. This limits the precision with which a pre-handover system can adapt its intervention.

Research into biometric cockpit sensing is active at major Tier-1 suppliers. Valeo and Continental have published work on systems that monitor heart rate variability (HRV), electrodermal activity (EDA), and facial dynamics to infer attentional states [16]. Recent HCI research has documented cockpit concepts that initiate protective responses, includ-

ing routing toward a hospital, when drowsiness is detected via infrared eye-movement scanning [7].

HABS AI, a Paris-Saclay startup, claims the capacity to detect more than 20 cognitive and emotional states, including mental workload and engagement level, from brainwave activity in real time, non-intrusively, using a headrest-integrated sensor architecture. In 2025, HABS formalised a research partnership with Université Paris-Saclay focused on cognitive technologies and applied field development [11].

The relevance to the pre-handover problem is direct. A system with access to real-time cognitive state could extend the preparation window when workload is high, compress it when the driver is already alert, and calibrate the visual transition sequence to match the driver’s actual recovery trajectory rather than a fixed time budget.

AI Confidence Communication

Research on AI confidence calibration shows that when AI systems display confidence levels, for example through a softmax probability derived from the final output layer, this shapes users’ willingness to accept or override the system [21]:

$$\text{Confidence} = \max_i(\sigma(z)_i) \quad (1)$$

where $\sigma(z)_i$ is the softmax function applied to output logits.

A study on confidence alignment found that human users tend to mirror the AI’s expressed confidence in their own self-confidence: a system expressing high certainty produces a calmer, more confident user, while a system expressing uncertainty produces an anxious, less capable one [1]. Real-time correctness feedback can reduce this alignment effect, helping users maintain a more accurate mental model of actual system performance [1].

For the pre-handover interface, the design implication is concrete. The visual language of the transition sequence should communicate that the system has the situation in hand, even as it alerts the driver to the approaching transition. An interface that signals uncertainty or urgency in the pre-handover window will elevate the driver’s stress at precisely the moment when the objective is the opposite.

Ethical and Regulatory Dimensions

Liability Transfer and Interface Responsibility

The L3 handover creates a specific and novel liability structure that is directly relevant to interface design decisions. When DRIVE PILOT is active, Mercedes-Benz accepts full legal liability for the vehicle’s operation [18]. The moment the TOR is issued, liability

begins transferring back to the driver. That transfer is legally contingent on the driver being capable of resuming control within the specified time budget.

This raises a concrete design question: if the pre-handover interface is absent, poorly timed, or poorly designed, and the driver cannot meet that time budget, where does responsibility lie? As L3 systems proliferate, the quality of the transition interface becomes a question with regulatory and legal implications, not just a UX concern. Interface designers working in this space are in practice making decisions that affect liability, and that context should shape how the pre-handover window is specified and evaluated.

Attention Manipulation and Consent

A pre-handover sequence that gradually interrupts entertainment and reintroduces driving context is, by design, a system that manipulates attention without explicit moment-to-moment consent. The driver consented to the system's overall behavior when they activated L3 mode, but they did not necessarily choose the specific timing, pacing, or form of the transition. EU guidelines for trustworthy AI require that systems affecting users be transparent and accountable, and that they respect human autonomy [8]. For a pre-handover interface, this translates into a set of concrete design requirements.

The system should communicate what it is doing and why. A visual transition sequence that dims entertainment content without any indication of the reason is less transparent than one that surfaces a brief status message. Drivers should be able to configure transition preferences where safety permits, for example choosing between an earlier gradual ramp and a shorter more decisive interruption. And any override mechanism must be legible without creating a path to unsafe behavior, a non-trivial design challenge in a safety-critical system.

These are not abstract ethical concerns. They are design constraints that affect how the Awareness Ramp and Re-engagement states are specified in practice.

Biometric Data and GDPR

If the pre-handover system incorporates cognitive state monitoring through HRV, EDA, or neural sensing technologies such as those under development at HABS, it generates sensitive personal data under the EU's General Data Protection Regulation [9]. The questions of who owns this data, whether it is stored onboard or transmitted to the OEM, and whether it is accessible to insurers or regulators are not peripheral concerns. They are design constraints that must be addressed in the system specification before deployment, not as an afterthought.

The interaction between L3 liability structures and biometric cockpit data is particularly worth flagging for future regulatory attention. If a driver's biometric state at the time of

a handover is recorded, that data could become relevant in any subsequent legal dispute about the adequacy of the driver's response. Designers and researchers working on cognitive state sensing in this context should be aware of that possibility and build appropriate data minimization and retention policies into the system from the outset.

Synthesis: Three Gaps and a Proposed Interface Architecture

Three Persistent Gaps

Drawing together the evidence from the preceding sections, three gaps in current L3 HMI design stand out:

Gap 1: The Silent AI. Current HMI is event-driven: the vehicle communicates only when something requires action. There is no ambient, continuous signal of the AI's operational state between events. The driver in autonomous mode has no sense of how the system is performing right now, whether it is operating well within its ODD or approaching its limits. This silence makes the TOR alert more shocking than it needs to be, because it arrives as the first signal of any kind that the driving situation is changing.

Gap 2: No Design Language for Uncertainty. There is a well-established design vocabulary for danger: red, loud, urgent. But there is no standardized vocabulary for communicating degrees of AI confidence or proximity to operational limits. The vehicle's automation is always operating on probability distributions, yet this internal state is communicated to the driver only when it crosses an emergency threshold. The question of what 60% system confidence should look or feel like compared to 90%, and whether drivers should even be exposed to such information, is entirely unaddressed in current HMI design.

Gap 3: The Undesigned Pre-Handover Window. Research and design effort concentrates almost entirely on the four-to-ten-second handover moment. The 30 to 60 seconds before that moment, the period during which situation awareness could be progressively rebuilt, physical reorientation initiated, and cognitive readiness prepared, is unaddressed in both the academic literature and in production systems. This is where the stress problem originates, and where the most promising design space lies.

A Five-State Interface Architecture

Based on the synthesis of the research literature, this paper proposes a five-state screen interface architecture for the L3 autonomous mode and handover transition. The five states are defined by their timing relative to the TOR and by their specific design function. Table 2 presents the architecture.

Table 2. Five-state interface architecture for L3 handover transition.

State	Timing	Interface content	Design challenge
Autonomous Mode	$T_{-\infty}$ to T_{-60s}	Entertainment unlocked. Passive AI status indicator present at screen periphery, continuous, non-intrusive, always visible	Ambient without distraction
Awareness Ramp	T_{-60s} to T_{-30s}	Entertainment dims progressively. Road preview appears at screen edge: current speed, lane position, surrounding vehicle count. Soft audio tone initiates	Interrupt without startle
Re-engagement	T_{-30s} to T_{-10s}	Entertainment pauses. Full road context displayed. Countdown visible. Driver prompted to return physical attention to driving position	Urgency without panic
Handover Alert	T_{-10s} to T_0	Full TOR. All channels active: visual, acoustic, haptic. Road situation displayed at maximum clarity. Required action unambiguous	Maximum clarity, zero clutter
Takeover Confirm	T_{+0} onward	Driver has control. Confirmation state displayed. AI status shows standby. Road context remains visible as stabilizing reference	Calm reassurance

The architecture is grounded in specific findings from the literature.

The **Autonomous Mode** ambient indicator addresses Gap 1 by establishing a continuous, low-salience signal of AI operational state that the driver can monitor passively. Its peripheral screen placement avoids visual competition with NDRT content, and its presence means the TOR no longer arrives as the first indication that anything is happening.

The **Awareness Ramp** draws directly on van der Heiden et al.’s finding that pre-alerts cause drivers to look at the road significantly more and disengage from secondary tasks earlier [33]. The progressive dimming of entertainment content follows the principle of interruption at natural task boundaries where possible [15]. The road preview at the screen edge implements the visual assistance approach shown in the 2024 Human Factors study to improve situation awareness at takeover [14].

The **Re-engagement** state is where the physical reorientation process identified by Zhao et al. and the Accident Analysis and Prevention posture study needs to begin [38, 30]. Displaying full road context at this stage also addresses cognitive residue: 30 seconds of road-context exposure before the TOR fires exceeds the 27-second decay window identified by Strayer et al. [32].

The **Handover Alert** state retains the multimodal urgency established in the literature as optimal for the final TOR [26, 28], while arriving to a driver whose attention has already been partially prepared by the preceding states.

The **Takeover Confirm** state addresses the confidence alignment effect noted in Section 5. A calm, confirmatory interface signal after control transfer helps stabilize the driver’s physiological and cognitive state at the moment it is most needed [1].

Adaptive Timing

The fixed timings in Table 2 represent the baseline case: a planned trigger with a full 60-second preparation window. For semi-predictable triggers (10–30 seconds), the Awareness Ramp and Re-engagement states compress or merge. For abrupt triggers (less than 10 seconds), the system proceeds directly to Handover Alert.

Where cognitive state sensing is available, session duration and measured cognitive load should be used to scale the transition. Longer sessions and higher cognitive load extend the Awareness Ramp and Re-engagement states; shorter sessions with lower cognitive load compress them. This directly addresses the out-of-the-loop scaling effect documented in [12] and the dynamic timing recommendation from [33].

Discussion and Future Work

Scope and Limitations

The framework presented here is derived from existing research and production system analysis. It has not been validated through empirical user testing, and the five-state architecture is a design proposal, not a tested intervention.

The research on which the architecture is based was conducted predominantly in driving simulators. While meta-analyses have demonstrated that simulator findings on distraction and response time translate to on-road behavior [5, 13], specific effect sizes may differ in naturalistic driving conditions, particularly given the rare and unpredictable nature of real-world TOR events in practice.

The timing parameters are derived from a small number of studies and a specific experimental context. Optimal timing for each state transition requires empirical calibration across driver populations, NDRT types, and trigger categories. The ambient AI status indicator proposed for the Autonomous Mode state is novel and has no direct empirical validation; its design, including visual form, placement, and salience calibration, requires dedicated user research.

Directions for Future Research

Several research directions follow directly from this framework.

Timing validation. A simulator study varying the onset of the Awareness Ramp (30, 45, 60, 90 seconds before TOR) would establish optimal preparation window length for different NDRT types and session durations. The interaction between session duration and preparation window length has not been systematically studied.

Information hierarchy in the Awareness Ramp. What road context information is most valuable for situation awareness restoration? Current speed, surrounding vehicles, lane markings, road geometry, and estimated TOR cause are all candidates. Empirical testing is needed to establish priority, and the optimal hierarchy likely varies by trigger type: a weather-related handover calls for different road context than a geofence boundary transition.

Ambient AI status design. What visual form, shape, color, animation, size, and screen position, best communicates AI operational state at screen periphery without drawing sustained visual attention away from entertainment content? This is a novel interaction design problem with no direct prior art. Peripheral display design research from other domains (ambient information displays, heads-up display research) provides relevant starting points, but the automotive L3 context introduces constraints those bodies of work did not address.

Individual differences and configurable profiles. Van der Heiden et al. found participant preferences for pre-alert type were split exactly 50/50 [33]. User research is needed on individual variability in transition preferences, and on whether configurable transition profiles are both acceptable to users and feasible within a safety-certified system. Some drivers may prefer an earlier, more gradual ramp; others may prefer a shorter, more decisive interruption. The safety implications of configurable timing require careful evaluation before any deployment.

Long-term habituation. Most research in this space is based on brief exposures, often a single simulator session. Recent HCI research notes that repetitive exposure to haptic stimuli reduces the annoyance threshold over time [7]. The same risk applies to any repeated transition sequence: over weeks and months of use, the interface must remain effective and non-intrusive. Whether progressive dimming of entertainment content retains its urgency signal after hundreds of exposures is an open question that cannot be answered without longitudinal study.

Evaluation methodology. Studying pre-handover interface design longitudinally poses methodological challenges. Simulator studies can test initial responses, but not habituation. Naturalistic driving studies can track real-world behavior, but TOR events in L3 systems are rare by design. Developing evaluation methods that can assess pre-handover interface quality across both short-term and long-term use is itself a research problem worth addressing.

Conclusion

Current L3 production systems handle the handover as an alarm problem. The driver is given four to ten seconds to respond to a simultaneous multimodal alert and resume full manual control from a cognitively and physically disengaged state. The research literature shows consistently that this is not enough.

The argument this paper makes is that the real design problem is not the alert. It is the 30 to 60 seconds before the alert fires. That is the window in which situation awareness can be rebuilt, physical reorientation can begin, and cognitive residue from the secondary task can decay. A screen-based transition sequence that begins the return process early, communicates road context progressively, and delivers the final TOR to a driver who is already partially ready represents a genuinely different approach from the current alarm paradigm.

The five-state architecture proposed here provides a concrete framework for that design, grounded in the existing empirical literature. It requires validation, and specific directions for that validation have been identified. The design space is real, the supporting evidence is there, and the production infrastructure to implement it already exists in deployed L3

vehicles.

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